

Lemma. An orthogonal set of non-zero vectors is linearly independent.

Proof. Let $S \subseteq V \setminus \{0\}$ be an orthogonal set.

Suppose that $\alpha_1, \dots, \alpha_n \in S$ are distinct vectors s.t.

$$\beta = c_1 \alpha_1 + \dots + c_n \alpha_n \text{ for some } c_i \in F.$$

Then, for each $1 \leq k \leq n$,

$$\langle \beta, \alpha_k \rangle = \langle \sum c_i \alpha_i, \alpha_k \rangle = \sum c_i \langle \alpha_i, \alpha_k \rangle = c_k \langle \alpha_k, \alpha_k \rangle \Rightarrow c_k = \frac{\langle \beta, \alpha_k \rangle}{\langle \alpha_k, \alpha_k \rangle}$$

Since each $\alpha_k \neq 0$, if $\beta = 0$, then $c_1 = \dots = c_n = 0$.

Furthermore, if $\beta \in V$ is linear combination of $\alpha_1, \dots, \alpha_n \in S$,

$$\beta = \sum_{i=1}^n \frac{\langle \beta, \alpha_i \rangle}{\langle \alpha_i, \alpha_i \rangle} \cdot \alpha_i.$$

Gram-Schmidt process.

Let V be a fin. dim. inn. prod. space, let $\{\beta_1, \dots, \beta_n\} \subseteq V$ be a linearly independent subset.

Then, there exists an orthogonal subset $\{\alpha_1, \dots, \alpha_n\} \subseteq V$ s.t.

$$\text{Span}\{\alpha_1, \dots, \alpha_n\} = \text{Span}\{\beta_1, \dots, \beta_n\}.$$

Proof. Using mathematical induction for n . For $n=1$, take $\alpha_1 = \beta_1$, then clear.

Now, suppose that for $m \in \mathbb{N}$, the statement holds. Let $\{\beta_1, \dots, \beta_{m+1}\}$ be lin. independent, and

$$\alpha_{m+1} = \beta_{m+1} - \sum_{i=1}^m \frac{\langle \beta_{m+1}, \alpha_i \rangle}{\langle \alpha_i, \alpha_i \rangle} \alpha_i \text{ where } \alpha_1, \dots, \alpha_m \text{ are obtained by hypothesis.}$$

Then $\alpha_{m+1} \neq 0$, because β_{m+1} is independent with $\{\alpha_1, \dots, \alpha_m\}$. Moreover, for $1 \leq j \leq m$,

$$\langle \alpha_{m+1}, \alpha_j \rangle = \langle \beta_{m+1}, \alpha_j \rangle - \sum_{i=1}^m \frac{\langle \beta_{m+1}, \alpha_i \rangle}{\langle \alpha_i, \alpha_i \rangle} \cdot \langle \alpha_i, \alpha_j \rangle = \langle \beta_{m+1}, \alpha_j \rangle - \langle \beta_{m+1}, \alpha_j \rangle = 0.$$

Now, $\{\alpha_1, \dots, \alpha_{m+1}\}$ is orthogonal, non-zero set, thus linearly independent by the Lemma.

Moreover, it is subset of $\text{Span}\{\beta_1, \dots, \beta_{m+1}\}$, the replacement Theorem gives result.